

PSTAT 5LS Lab 6 Notes

Fall 2025

Lab Notes

Use this place to take any notes during your lab section.

- Type any notes here
- Add new notes by using the dash - to create a list

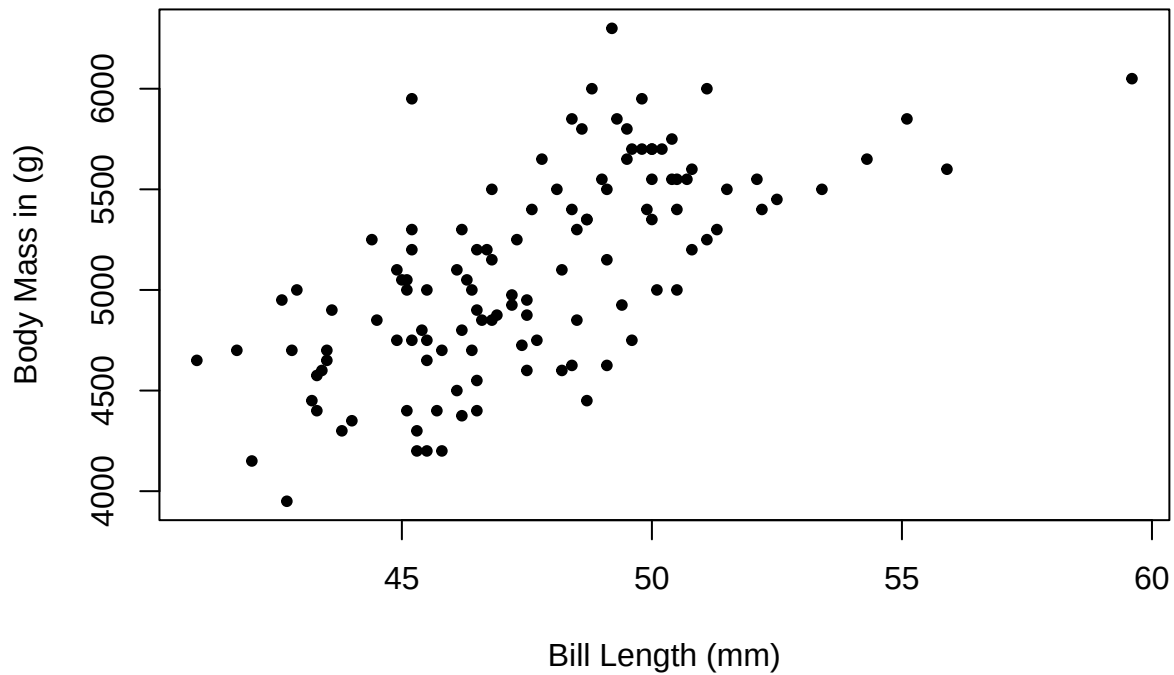
Lab Code

Use the chunks below to try out any code that you'd like.

```
# Run this code chunk to make the `gentoo` data set available for analysis  
gentoo <- read.csv("gentoo.csv", stringsAsFactors = TRUE)
```

```
# Run this code chunk to create a scatter plot with bill_length_mm as the  
# explanatory variable and body_mass_g as the response variable  
plot(body_mass_g ~ bill_length_mm,  
     data = gentoo,  
     main = "Scatter plot of Body Mass versus Bill Length for Gentoos",  
     xlab = "Bill Length (mm)",  
     ylab = "Body Mass in (g)",  
     pch = 20)
```

Scatter plot of Body Mass versus Bill Length for Gentoos



```
# Run this code chunk to compute the correlation between bill_length_mm and body_mass_g
cor(gentoo$bill_length_mm, gentoo$body_mass_g)
```

```
## [1] 0.6667302
```

```
# Run this code chunk to subset the `gentoo` data to contain only numeric variables. We will call this .
numericGentoo <- subset(gentoo,
  select = c("bill_length_mm", "bill_depth_mm",
             "flipper_length_mm", "body_mass_g")
)
```

```
# Run this code chunk to make a correlation matrix of `numericGentoo`
cor(numericGentoo)
```

```
##               bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
## bill_length_mm      1.0000000      0.6540233      0.6642052      0.6667302
## bill_depth_mm       0.6540233      1.0000000      0.7106422      0.7229672
## flipper_length_mm   0.6642052      0.7106422      1.0000000      0.7113053
## body_mass_g         0.6667302      0.7229672      0.7113053      1.0000000
```

Code Cheat Sheet

~

- Read as “by”
- Used in side-by-side boxplots, scatter plots, and finding linear models
- $y \sim x$ is the correct syntax for linear; `numeric_variable ~ categorical_variable` is the correct syntax for side-by-side boxplots

plot(*y ~ x*, data = data_name)

- Used to create a scatter plot of the *y* variable by the *x* variable
- Can use the argument **data = data_name** to eliminate the need to specify where each of the variables *x* and *y* are contained

cor(*x*, *y*)

- Finds the correlation coefficient between the numeric variable *x* and the numeric variable *y*

cor(numeric_data_frame)

- Prints a correlation matrix for a data frame with all numeric variables

Important plotting arguments

main = "Title of Your Graph in Double Quotes"

- graph title that must be inside a set of double quotes

xlab = "x-axis Label of Your Graph in Double Quotes"

- the x- (horizontal) axis label that must be inside a set of double quotes

ylab = "y-axis Label of Your Graph in Double Quotes"

- the y- (vertical) axis label that must be inside a set of double quotes